

IEEE GAME JAM

“You are the controller”

COMBO MANUAL

Pick your parts. Stick them together. Get a game. Every recipe below is one way to start. Steal it, twist it, mash two together, or invent your own.

DIFFICULTY

LEVEL 1

One type of sensor, and each input simply sends a key.

LEVEL 2

Two types of sensor, two players on one board, a sensor that needs tuning, or the camera.

LEVEL 3

Three or more types of sensor, where the ESP32 combines readings before sending a key.

KEY TRICK: The ESP32 acts like a keyboard, so it can only press keys. For “harder” or “closer” actions, send a different key for each strength, such as one key for a light stomp and another for a hard stomp.

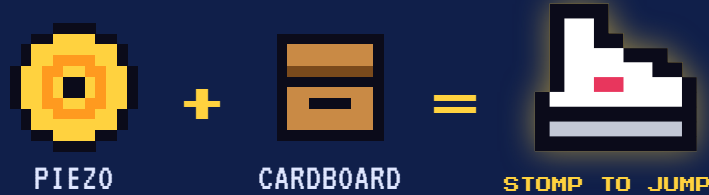
CRAFTING RECIPES

This + this = a game you can build today.

LEVEL 1

LAVA FLOOR JUMP

LEVEL 1



Hide a piezo disc under a cardboard foot pad. **Stomp** to make your character jump over lava. A harder stomp means a higher jump.

Keys: one for a light stomp, another for a hard stomp.

One more round: build two pads and race a friend to the top.

DRUM HERO

LEVEL 1



Tape piezo discs under four cardboard drum pads. Notes fall down the screen, and you **hit** the right pad as each note lands.

Keys: one per pad.

One more round: a combo counter that resets on every miss.

CLAP ROCKET

LEVEL 1



SOUND
SENSOR

=



CLAP TO FLY

Every **clap** or **shout** boosts your rocket upwards. Stop making noise and it falls. Steer it through asteroids, or through gaps between walls, without crashing.

Keys: one whenever the sound is loud enough.

Watch out: the room will be noisy. Keep the sensor close and turn its sensitivity down so other teams don't play your game for you.

WHACK-A-MOLE BOX

LEVEL 1



BUTTONS

+



CARDBOARD
BOX

=



WHACK-A-MOLE

Cut holes in a cardboard box with a button or limit switch under each one. Moles pop up on screen, and you **smack** the matching hole before they hide.

Keys: one per hole.

One more round: two players, two boxes, same moles. Fastest hands win.

QUICK DRAW

LEVEL 1



2x BUTTON

=



REACTION DUEL

Two players, one button each. The screen says "Wait...", then suddenly "DRAW!". First to **press** wins the round. Press too early and you lose it.

Keys: one per player.

One more round: first to five wins, with a random wait so no one can guess the timing.

BUTTON MASH TUG-OF-WAR

LEVEL 1



Two players **mash** their buttons as fast as they can. Every press pulls the rope one step towards you. Drag the other player over the line to win.

Keys: one per player.

One more round: a “power pull” bonus that appears for a second, worth five steps to whoever presses first.

DINO WAVE

LEVEL 1



Your character runs non-stop. **Wave** your hand over the IR sensor to jump over cacti and obstacles. Miss one and it's game over.

Keys: one whenever a hand passes over the sensor.

One more round: the game speeds up the longer you survive.

LEVEL 2

TURBO WHEEL

LEVEL 2



Build a cardboard steering wheel with the MPU-6050 inside. Tilt to steer. Put two limit switches under cardboard foot pedals for nitro and brakes.

Keys: ← and → for steering, plus one per pedal.

One more round: a lap timer and a best-time board.

HOVER HAND

LEVEL 2



Your hand over the ultrasonic sensor works like a joystick. Keep it in the middle to hover, and move it up or down to fly. The further you move it, the faster you climb or drop.

Keys: five, from closest to furthest: down a lot, down a little, hover, up a little, up a lot.

Keep it close: the sensor is only reliable up to about 30 cm, so keep all five positions within that range.

TILT TANK TUSSLE

LEVEL 2



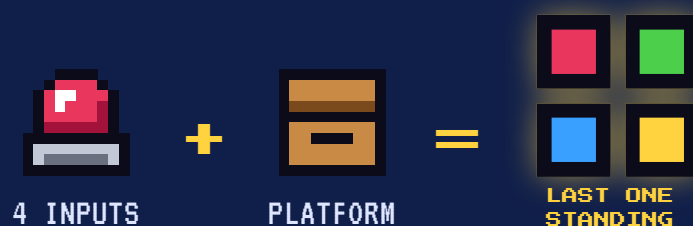
Each player gets a cardboard tank with an MPU-6050 inside. Tilt to steer, press to fire. Head-to-head on one screen.

Keys: each player gets their own set, such as W A S D + Space, and the arrow keys + Enter.

Wiring: both sensors can share one ESP32. Connect the second sensor's AD0 pin to 3.3 V so the board can tell them apart.

SIMON SAYS: BATTLE ROYALE

LEVEL 2



The screen flashes a sequence of four colours. Every player repeats it on their own input platform. Miss one and you're out. The sequence grows each round.

Inputs: any mix works, such as two buttons, a piezo pad to slap, and an IR sensor to wave over.

Keys: one per input. Up to 4 players can share one ESP32, each with their own four keys.

BODY DODGE

LEVEL 2



LAPTOP
CAMERA

=



DODGE THE
BLOCKS

No ESP32 needed. The laptop's camera tracks your body, and your character copies you. **Duck and dodge** the blocks falling from the sky.

How: a web game using MediaPipe, a free Google tool that finds your body or hands in a webcam video, right in the browser.

One more round: blocks fall faster every 10 seconds.

LEVEL 3

RAGE QUIT SIMULATOR

LEVEL 3



MPU-6050

+



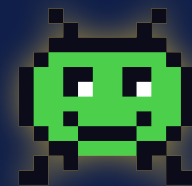
PIEZO

+



SOUND
SENSOR

=



LASER
SHOWDOWN

Shake the controller to charge a laser. **Slam** the table to fire. **Yell** to raise a shield.

Keys: one for shaking, one for a slam, and one for a shout.

Detecting a shake: a shake shows up as big, fast jumps in the MPU-6050's reading. Send the shake key when it jumps past a set limit.

BOMB DEFUSER

LEVEL 3



ULTRASONIC

+



MPU-6050

+



PIEZO

+



SOUND
SENSOR

+



BUTTONS

+



BOTTLE

=



CO-OP
DEFUSING

The bomb is a plastic bottle covered in sensor modules. Before the timer runs out, the screen tells you to: hold your hand exactly 15 cm over the **ultrasonic** sensor, keep the bottle perfectly **level**, **tap** a code on the piezo, **shout** to cool it down, and **press** the buttons in order. Best in pairs: one handles the bomb, one reads the screen.

Keys: one per module when it's solved. The ESP32 checks each module, so the game only listens for "solved".

One more round: a shorter timer and more modules every level.

WIZARD DUEL

LEVEL 3



2x MPU-6050

+



2x IR
SENSOR

+



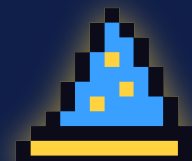
SOUND
SENSOR

+



2x BOTTLE

=



SPELL BATTLE

Each wand is a plastic bottle with an MPU-6050 inside. **Flick** it up, down, left, or right to cast four spells. **Wave** over your IR sensor to block. **Shout** to power up your next spell.

Keys: four spell keys and one block key per player, plus a power-up key.

Detecting a flick: check which direction had the biggest jump in the reading, and send that spell's key. Both wands share one ESP32 (second sensor's AD0 pin to 3.3 V).

CO-OP STARSHIP

LEVEL 3



MPU-6050

+



BUTTONS

+



IR SENSOR

+



SOUND
SENSOR

+



BOTTLE

=



CO-OP
SPACESHIP

Two players fly one ship, and neither can survive alone. The **pilot** steers with an MPU-6050 inside a plastic bottle. The **gunner** fires with the buttons, **waves** over the IR sensor to reload, and **shouts** to raise the shields.

Keys: ← and → for steering, plus one each for firing, reloading, and shields.

One more round: enemy waves that get tougher each level, with a team high score.

MAKE YOUR OWN

1 **INPUT** (tilt / tap / stomp / clap / wave / press)
+ 1 **TWIST** (second sensor, second player, or camera)
+ 1 **REASON TO PLAY AGAIN** (score, timer, or a friend to beat)
= **YOUR GAME**

MAKE IT FEEL ARCADE

SHORT ROUNDS

Under two minutes, so the next player gets a go.

SCORE ON SCREEN

Always. People play again to beat a number.

INSTANT RESTART

One button, straight back in.

SOUND = 50%

A beep on every hit makes any game feel ten times better.

TWO PLAYERS BEAT ONE

Competing against a friend keeps the queue going.

PLAY TO THE CROWD

Big text and bright colours, so the people waiting can watch too.

GAME OVER? INSERT COIN TO CONTINUE

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